The long-root collapse multiplier is type-dependent,

and it is the square of the charge-violating boundary entry
$\rho^X - 1 = q\, w_a^h \big([K^X(w_a)]_0^1\big)^2$, $\quad h = 2$ in type C, $h = 1$ in type B;
a retraction of the type B claim of `2026-10-16-c2`, and a reduction of Gap `nonvanH` to $q = 0$

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PROTOCOL §2.3 record.

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Abstract

This continues `proofs/2026-10-16-c2-longroot-multiplier.tex` (henceforth [LR]) and corrects it.

[LR] Theorem `collapse` states that on a long-root divisor $H_{ab} = \{w_a w_b = 1\}$ Motegi's normalised weight functions (arXiv:2610.15344) degenerate by a universal scalar $\rho = (1-q)^2/\big((1-qw_a^2)(1-qw_a^{-2})\big)$, and asserts that "*the same statement and the same ρ hold in type B*", verified 48/48. **The multiplier claim is false and is retracted here.** The verification measured $WB/\mathcal{E}^C$: my engine divided the type B partition function by the type C normalisation factor. Motegi defines $\widetilde{WB} := WB \cdot (\mathcal{E}^B)^{-1}$ with $\mathcal{E}^B \neq \mathcal{E}^C$, and

$$\mathcal{E}^B/\mathcal{E}^C\big|_{\mathbf{u}=\mathbf{w_M}} = \prod_{k\notin\mathbf{M}} \frac{1-qw_k^2}{1-qw_k}$$

(Lemma 1), whose $\mathbf{M}$-dependence does not cancel in a collapse ratio. The correct type B multiplier is

$$\rho^B = \frac{(1-q)^2}{(1-qw_a)(1-qw_b)}, \qquad \text{against} \qquad \rho^C = \frac{(1-q)^2}{(1-qw_a^2)(1-qw_b^2)}$$

(Theorem 3): the *long*-root factors $1 - qe^{+2t_c}$ of type C are replaced by the *short*-root factors $1 - qe^{+t_c}$. Each candidate holds on its own type and fails on the other, over a domain on which they disagree everywhere.

The correction is not a loss: it supplies the discriminating coordinate that makes the boundary reading a statement rather than a resemblance. Writing $[K^X(u)]_0^1$ for the charge-violating entry of Motegi's triangular K-matrix — $1-u^{-2}$ in type C, $1-u^{-1}$ in type B — Theorem 6 is the identity

$$\rho^X - 1 = \frac{q\, w_a^h \big([K^X(w_a)]_0^1\big)^2}{(1-qw_a^h)(1-qw_b^h)}, \qquad h = 2\ (X = C), \quad h = 1\ (X = B),$$

with h the length of the boundary root. The naive form of this reading uses the *unnormalised* entry, which is β,γ-free hence identical in both types, and therefore predicts a type-independent ρ: that form is refuted by Theorem 3. The two readings agree at $w_a = +1$ and disagree at $w_a = -1$, and the measurement lands on the disagreement set. The slot is also pinned, and from a source theorem rather than from my own algebra: Motegi's divisibility theorems make $\prod_{m\in\mathbf{M}}(1 - w_m^{-h})$ a divisor of $\widetilde{WX}_{\mathbf{M},\mathbf{I}}$, indexed by the specialisation set $\mathbf{M}$ — which is exactly the index set of the K-matrices, since $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ is the specialisation $\mathbf{u} = \mathbf{w}_{\mathbf{M}}$ and K acts on the $\mathbf{u}$-slot. A long-root collapse deletes $\{a, b\}$ from $\mathbf{M}$, hence deletes exactly the two K-matrices $K(w_a)$, $K(w_b)$.

On the primary target, Gap `nonvanH` ($\mathrm{ord}_{H_{ab}}\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ when $\mathbf{M} \geq \mathbf{I}$ and $\{a, b\} \cap \mathbf{M} = \emptyset$), I report a reduction and two failed attempts with their breakdown points. The reduction (Proposition 10) is that it suffices to prove the statement at $q = 0$, because $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ is regular at $q = 0$ and $\mathcal{E}^C|_{q=0} = 1$; and that at $q = 0$ the entire dependence of $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ on the variables outside $\mathbf{w}_{\mathbf{M}}$ is the single product $\prod_{j\leq r}\prod_{k<i_j}(u_j - w_k)$, so that w_k is absent whenever $k \geq \max\mathbf{I}$. This closes the gap on the stratum $\max\mathbf{I} \leq b$ — 100%, 91.7%, 81.9%, 72.8%, 64.8%, 58.0%, 52.3% of the domain at $N = 2, \ldots, 8$, *decreasing* — modulo nonvanishing of $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ at $q = 0$, which is a factorial *GQ*-function statement and is *stronger* than the converse of triangularity that [Q488] already carries as measured. I say so rather than calling it a closure. The two failed attempts are recorded in §5.3: a bidegree-corner argument, *refuted by measurement* (the corner is empty in 12/12 cases at $N = 3$), and its $q = 0$ refinement, which reduces to a symmetrisation sum twisted by a non-symmetric monomial that I cannot evaluate. Gap `nonvanH` remains open.

Everything [LR] states for type C is untouched: $\mathcal{E}^C$ is the correct normaliser for type C and is checked against Motegi's independent diagonal root-product formula.

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1 What this corrects, and how the error was found

The session brief armed a gate at position zero: a dictionary, to be written by hand, deciding whether the variable w_a appearing in

$$\rho - 1 = \frac{q\,(w_a - w_a^{-1})^2}{(1 - qw_a^2)(1 - qw_a^{-2})} \qquad \text{([LR] Prop. rho)}$$

occupies the same slot as the argument u of Motegi's boundary K-matrix. The brief's own pre-registered worry was the right one — "same symbol, possibly different slot" — and the gate paid for itself three times over, in ways the brief did not anticipate.

First: the brief quoted the wrong paper. The brief records, as verified verbatim at wake, ${}_a\langle 1|K_a(u)|0\rangle_a = u - u^{-1}$ from `arxiv-1703.07924-source.TEX`, eq. `kmatrix`, L300–321. That citation is accurate, and it is about a different paper. The theorem under discussion is about arXiv:2610.15344, whose triangular K-matrix carries the *opposite* triangularity (eq. `eq:Ktri-components`, L332–345):

$$[K(u)]_i^j = \begin{cases} \beta - \gamma u^{-1} & (i,j) = (0,0), \\ 1 - u^{-2} & (i,j) = (0,1), \\ 0 & (i,j) = (1,0), \\ \beta u^{-2} - \gamma u^{-1} & (i,j) = (1,1). \end{cases} \tag{1}$$

In 1703.07924 the charge-violating entry is the $(1,0)$ entry and equals $u - u^{-1}$; here the $(1,0)$ entry is *zero* and the charge-violating entry is the $(0,1)$ entry, $1 - u^{-2}$. The two are proportional, $u - u^{-1} = u\,(1 - u^{-2})$, so the character-level match that excited the brief survives — but only one of the two is a quantity of the paper being proved about, and the discrepancy by a factor of u is exactly the monomial w_a^h that appears in Theorem 6. Two Motegi papers, the same phrase "triangular K-matrix", transposed conventions.

Second: the two specialisations of (1). Motegi specialises (1) twice (eqs. `eq:KC-components`, `eq:KB-components`, `triandc`, `triandb`, L352–375):

$$K^C(u) = K(u)\big|_{\gamma=0,\beta=1}, \qquad K^B(u) = (1+u^{-1})^{-1} K(u)\big|_{\gamma=-1,\beta=1}, \tag{2}$$

so that the charge-violating entries are

$$[K^C(u)]_0^1 = 1 - u^{-2}, \qquad [K^B(u)]_0^1 = 1 - u^{-1}. \tag{3}$$

Note that in (1) the $(0,1)$ entry does not involve β or γ at all. So the *unnormalised* charge-violating entry is literally the same function in both types, and only the type B scalar $(1+u^{-1})^{-1}$ of (2) distinguishes them. This is the fork on which the whole reading turns, and §4 resolves it.

Third: the engine was normalising type B by $\mathcal{E}^C$. Checking the slot required reading Motegi's definitions of the two normalisation factors, and they are not the same function. That is §2. It invalidates every type B number in [LR] and in [Q484], and it changes a theorem.

Notation. As in [LR]. $\mathbb{F} = \mathbb{C}(q)(w_1, \ldots, w_N)$; for $\mathbf{M}, \mathbf{I} \subseteq [N]$ with $|\mathbf{I}| = r$ and $|\mathbf{M}| = s \geq r$,

$$\widetilde{A}^X_{\mathbf{M},\mathbf{I}} := \widetilde{WX}_{N,s,r}(\mathbf{w}_{\mathbf{M}} \,|\, \mathbf{w} \,|\, \mathbf{I}), \qquad X \in \{B, C\}, \tag{4}$$

and $\widetilde{A}_{\mathbf{M},\mathbf{I}} := \widetilde{A}^C_{\mathbf{M},\mathbf{I}}$. Throughout $a < b$ are fixed, $H := H_{ab} = \{w_a w_b = 1\}$, $t := w_a w_b - 1$, and $f \mapsto f|_H$ is restriction (i.e. the substitution $w_b \mapsto w_a^{-1}$). Motegi's order $\mathbf{M} \geq \mathbf{I}$ is as restated in [LR] Lemma `order`. All line numbers refer to the e-print source `typebcweightfunctions_arxivversionfinal-3.tex` of arXiv:2610.15344 unless the file is named otherwise.

2 The two normalisation factors

Motegi's unnormalised weight functions are two specialisations of one partition function $F_{N,n,r}(\mathbf{u}|\mathbf{w}|\mathbf{I})$ (eqs. `WCfunction`, `WBfunction`, L489–505):

$$WC = F\big|_{\gamma=0,\beta=1}, \qquad WB = \Big(\prod_{k=1}^{n}(1+u_k^{-1})^{-1}\Big)F\big|_{\gamma=-1,\beta=1}, \tag{5}$$

and the normalised ones are $\widetilde{WX} := WX \cdot (\mathcal{E}^X)^{-1}$ (eqs. `typeCnormalizeddefinition` L1051, `typeBnormalizeddefinition` L1358), with

$$\mathcal{E}^C_{N,n,r} = \frac{1}{(1-q)^{n-r}}\,\frac{\prod_{j,k=1}^{n}(1-qu_j^{-1}u_k)\ \prod_{j=1}^{n}\prod_{k=1}^{N}(1-qu_jw_k)}{\prod_{1\le j<k\le n}(1-qu_ju_k)\ \prod_{1\le j\le k\le N}(1-qw_jw_k)}, \tag{6}$$

$$\mathcal{E}^B_{N,n,r} = \frac{1}{(1-q)^{n-r}}\,\frac{\prod_{j,k=1}^{n}(1-qu_j^{-1}u_k)\ \prod_{j=1}^{n}\prod_{k=1}^{N}(1-qu_jw_k)\ \prod_{j=1}^{n}(1-qu_j)}{\prod_{1\le j<k\le n}(1-qu_ju_k)\ \prod_{j=1}^{n}(1-qu_j^2)\ \prod_{1\le j<k\le N}(1-qw_jw_k)\ \prod_{k=1}^{N}(1-qw_k)} \tag{7}$$

(eqs. `typeCnormalizationfactor` L1057–1062, `typeBnormalizationfactor` L1361–1367).

The difference is not cosmetic and it is not a scalar. Read the two denominators: $\mathcal{E}^C$ carries the $\mathbf{w}$-product over $j \le k$, i.e. *including* the diagonal $\prod_k(1-qw_k^2)$, and no $\mathbf{u}$-diagonal; $\mathcal{E}^B$ carries the $\mathbf{w}$-product over $j < k$ *strictly*, has the diagonal moved to the $\mathbf{u}$-side as $\prod_j(1-qu_j^2)$, and in its place carries the linear product $\prod_k(1-qw_k)$. Under the specialisation $\mathbf{u} = \mathbf{w}_\mathbf{M}$ the $\mathbf{u}$-diagonal becomes a $\mathbf{w}$-diagonal over $\mathbf{M}$, and the two conventions then differ exactly on the complement.

Lemma 1 (the ratio of the normalisers)**.** *For $\mathbf{M} \subseteq [N]$ with $|\mathbf{M}| = n$, and any $r \le n$,*

$$\frac{\mathcal{E}^B_{N,n,r}}{\mathcal{E}^C_{N,n,r}}\bigg|_{\mathbf{u}=\mathbf{w}_\mathbf{M}} = \prod_{k\in[N]\setminus\mathbf{M}}\frac{1-qw_k^2}{1-qw_k}. \tag{8}$$

Equivalently, with $G_\mathbf{M}$ denoting the right-hand side, $\widetilde{A}^B_{\mathbf{M},\mathbf{I}} = \big(WB/\mathcal{E}^C\big)\big|_{\mathbf{u}=\mathbf{w}_\mathbf{M}} \cdot G_\mathbf{M}^{-1}$.

Proof. Every factor of (6) occurs in (7), so the quotient is

$$\frac{\mathcal{E}^B}{\mathcal{E}^C} = \frac{\prod_{j=1}^{n}(1-qu_j)}{\prod_{j=1}^{n}(1-qu_j^2)}\cdot\frac{\prod_{1\le j\le k\le N}(1-qw_jw_k)}{\prod_{1\le j<k\le N}(1-qw_jw_k)\ \prod_{k=1}^{N}(1-qw_k)} = \prod_{j=1}^{n}\frac{1-qu_j}{1-qu_j^2}\cdot\prod_{k=1}^{N}\frac{1-qw_k^2}{1-qw_k},$$

the second equality because the two $\mathbf{w}$-products over pairs cancel except on the diagonal. Substituting $\mathbf{u} = \mathbf{w}_\mathbf{M}$ makes the first product $\prod_{m\in\mathbf{M}}\frac{1-qw_m}{1-qw_m^2}$, which cancels the $\mathbf{M}$-part of the second, leaving (8). Note that no factor simplifies further: $1-qu^2$ is not $(1-qu)(1+qu)$, the q being to the first power in each. □

Remark 2 (the swap is the long/short root swap). Lemma 1 is the statement that type C's ascent factor at the index k is $1-qw_k^2 = 1-q\,e^{+2t_k}$ and type B's is $1-qw_k = 1-q\,e^{+t_k}$: the *long* root $2t_k$ of C_N against the *short* root t_k of B_N. This is corroborated, independently of (6)–(7), by Motegi's two divisibility theorems (eqs. `typeCfullDivisibility` L1086–1089 and `typeBfullDivisibility` L1394–1402), which state that for $|\mathbf{M}| = |\mathbf{I}| = r$ the function $\widetilde{WX}_{\mathbf{M},\mathbf{I}}$ is divisible by

	$\mathbf{M}$-side (inversion)	$\mathbf{M}^{\mathrm{c}}$-side (ascent)
$X = C$	$\prod_{1\le j\le k\le r}(1 - w_{m_j}^{-1}w_{m_k}^{-1})$	$\prod_{j\le k\in\mathbf{M}^{\mathrm{c}}}(1 - qw_jw_k)$
$X = B$	$\prod_j(1 - w_{m_j}^{-1})\prod_{j<k}(1 - w_{m_j}^{-1}w_{m_k}^{-1})$	$\prod_{j\in\mathbf{M}^{\mathrm{c}}}(1 - qw_j)\prod_{j<k\in\mathbf{M}^{\mathrm{c}}}(1 - qw_jw_k)$

(both also carry $\prod_{j\in\mathbf{M}^{\mathrm{c}}}\prod_{l:m_l<j}(1 - qw_{m_l}^{-1}w_j)$). The diagonals are $1 - w_m^{-2}$ versus $1 - w_m^{-1}$ on the $\mathbf{M}$-side and $1 - qw_j^2$ versus $1 - qw_j$ on the complement: the same swap, on both sides, in a theorem that is not a normalisation factor. This is what §4 consumes, and it is how the implementation of (7) is checked (§6).

3 The type B collapse multiplier

Retraction 1. *[LR] Theorem* `collapse` *ends with: "*The same statement and the same ρ hold in type B*", and [LR] §*`verif` *reports* 48/48 *at $N = 3$ for type B with* 12 *non-degenerate. The* statement *survives; the words "the same ρ" are withdrawn. The computation behind the* 48/48 *divided WB by $\mathcal{E}^C$ — the engine* `q484_engine.Atilde` *carries an in-code comment asserting that "$\mathcal{E}^C$ is the common normalizer", which is false — so it is a correct measurement of $WB/\mathcal{E}^C$ and not of $\widetilde{WB} = WB/\mathcal{E}^B$. By the same token the type B arm of [Q484] Theorem A (*48/48*) measures the same wrong object and is withdrawn pending re-measurement. Nothing stated for type C is affected: $\mathcal{E}^C$ is the correct normaliser there.*

Theorem 3 (the type B multiplier)**.** *Let $a < b$, $\{a, b\} \subseteq \mathbf{M}$, $\mathbf{M}' = \mathbf{M} \setminus \{a, b\}$. Then for every $\mathbf{I} \subseteq [N]$,*

$$\widetilde{A}^B_{\mathbf{M},\mathbf{I}} \equiv \rho^B \cdot \widetilde{A}^B_{\mathbf{M}',\mathbf{I}} \pmod{w_aw_b - 1}, \qquad \rho^B = \frac{(1-q)^2}{(1-qw_a)(1-qw_b)}, \tag{9}$$

and $\rho^B \neq \rho^C$, where $\rho^C = (1-q)^2/\big((1-qw_a^2)(1-qw_b^2)\big)$ is the multiplier of [LR] Theorem `collapse` *(there written with w_b^2 replaced by its value w_a^{-2} on H). Uniformly in the type,*

$$\rho^X = \frac{(1-q)^2}{(1-qw_a^h)(1-qw_b^h)}, \qquad h = 2\ (X = C), \quad h = 1\ (X = B), \tag{10}$$

with h the length of the boundary root: $2t_c$ in type C, t_c in type B.

Proof. Write $\Lambda_{\mathbf{M},\mathbf{I}} := \big(WB/\mathcal{E}^C\big)|_{\mathbf{u}=\mathbf{w}_\mathbf{M}}$ for the object [LR] measured, so that $\widetilde{A}^B_{\mathbf{M},\mathbf{I}} = \Lambda_{\mathbf{M},\mathbf{I}}\, G_\mathbf{M}^{-1}$ by Lemma 1, with $G_\mathbf{M} = \prod_{k\notin\mathbf{M}} \frac{1-qw_k^2}{1-qw_k}$. Then

$$\frac{\widetilde{A}^B_{\mathbf{M},\mathbf{I}}}{\widetilde{A}^B_{\mathbf{M}',\mathbf{I}}} = \frac{\Lambda_{\mathbf{M},\mathbf{I}}}{\Lambda_{\mathbf{M}',\mathbf{I}}} \cdot \frac{G_{\mathbf{M}'}}{G_\mathbf{M}}, \qquad \frac{G_{\mathbf{M}'}}{G_\mathbf{M}} = \prod_{k\in\mathbf{M}'^{\mathrm{c}}\setminus\mathbf{M}^{\mathrm{c}}} \frac{1-qw_k^2}{1-qw_k} = \frac{(1-qw_a^2)(1-qw_b^2)}{(1-qw_a)(1-qw_b)}, \tag{11}$$

since $\mathbf{M}'^{\mathrm{c}} = \mathbf{M}^{\mathrm{c}} \sqcup \{a, b\}$. [LR]'s type B measurement, reproduced here (§6), is exactly $\Lambda_{\mathbf{M},\mathbf{I}}|_H = \rho^C\, \Lambda_{\mathbf{M}',\mathbf{I}}|_H$, and the quotient (11) is a unit at H with no pole or zero there (each factor is $1 - q\cdot$(monomial), nonzero because q is an indeterminate). Substituting,

$$\widetilde{A}^B_{\mathbf{M},\mathbf{I}}\big|_H = \rho^C \cdot \frac{(1-qw_a^2)(1-qw_b^2)}{(1-qw_a)(1-qw_b)} \cdot \widetilde{A}^B_{\mathbf{M}',\mathbf{I}}\big|_H = \frac{(1-q)^2}{(1-qw_a)(1-qw_b)}\, \widetilde{A}^B_{\mathbf{M}',\mathbf{I}}\big|_H.$$

That $\rho^B \neq \rho^C$ in $\mathbb{F}$ is (11) together with $(1-qw_a^2)(1-qw_b^2) \neq (1-qw_a)(1-qw_b)$. □

Remark 4 (where the proof of [LR] Theorem `collapse` goes through, and where it does not). The proof in [LR] is a factor-by-factor comparison of the branching formula at $\mathbf{M}$ and at $\mathbf{M}'$, and its final step is: "*what survives is the one factor of* Π_1 *with both legs in* $\{a,b\}$*, against the leading* $(1-q)^2$", where $\Pi_1 = \prod_{j\le k\in\mathbf{M}^c}(1-qw_jw_k)$ is the image in the numerator of $\widetilde{A}$ of the $\mathbf{w}$-product in the denominator of $\mathcal{E}^C$. In type B that product is $\prod_{j<k\in\mathbf{M}^c}(1-qw_jw_k)\cdot\prod_{j\in\mathbf{M}^c}(1-qw_j)$ — strictly off-diagonal, plus a linear product (Remark 2). So there *is* no factor with both legs in $\{a,b\}$ beyond $(1-qw_aw_b)$, which restricts to $1-q$ and is cancelled by Π_4's $(1-qw_a^{-1}w_b^{-1})|_H = 1-q$ exactly as in [LR]; and what survives instead is the pair of linear factors $(1-qw_a)(1-qw_b)$. The proof above routes around this by going through Λ rather than re-deriving the branching formula in type B, which is why it does not need the type B cross-sector branching formula — an input [LR] asserted ("the cross-sector branching formula has the same shape") without checking. The factor-by-factor reading and the Λ-route give the same answer, which is one of the two independent confirmations recorded in §6.

The sentence in [LR] that *was* right, and which now reads as the warning it should have been, is: "the type B specialisation differs from type C by $\beta=1,\gamma=-1$ and the prefactor $\prod_k(1+u_k^{-1})^{-1}$, neither of which contributes a factor vanishing on a long-root divisor." Both halves of that are true. Neither bears on the *value* of the multiplier, and the normalisation factor — the thing that actually differs — is not mentioned. A correct observation about vanishing was allowed to stand in for an unmade observation about value.

4 The dictionary: $\rho^X - 1$ is a boundary entry, squared

Proposition 5. *With* h *as in* (10)*, on* H *(where* $w_b = w_a^{-1}$*),*

$$\rho^X - 1 = \frac{q\left(w_a^{h/2} - w_a^{-h/2}\right)^2}{(1-qw_a^h)(1-qw_a^{-h})}, \tag{12}$$

so that $\operatorname{ord}_H \rho^X = \operatorname{ord}_H(\rho^X-1) = 0$ *and* $\rho^X \neq 1$*. Explicitly, the numerator is* $q(w_a - w_a^{-1})^2$ *in type* C *and* $q(w_a + w_a^{-1} - 2)$ *in type* B*.*

Proof. $(1-q)^2 - (1-qx)(1-qx^{-1}) = (1-2q+q^2) - (1-q(x+x^{-1})+q^2) = q(x+x^{-1}-2) = q(x^{1/2}-x^{-1/2})^2$ for any x; apply it with $x = w_a^h$, using $w_b^h = w_a^{-h}$ on H. Neither numerator nor denominator involves w_b, so neither is divisible by t. □

Theorem 6 (the boundary dictionary)**.** *For* $X \in \{B,C\}$*, with* $[K^X(u)]_0^1$ *the charge-violating entry* (3) *of Motegi's* K*-matrix and* $h = 2$ *resp.* 1*,*

$$[K^X(u)]_0^1 = u^{-h/2}\left(u^{h/2} - u^{-h/2}\right), \qquad \textit{hence} \qquad \rho^X - 1 = \frac{q\,w_a^h\left([K^X(w_a)]_0^1\right)^2}{(1-qw_a^h)(1-qw_b^h)}\bigg|_H. \tag{13}$$

In particular the zero locus in w_a *of the obstruction* $\rho^X - 1$ *equals the zero locus of the charge-violating boundary entry, in each type separately:* $\{w_a = \pm 1\}$ *in type* C *and* $\{w_a = +1\}$ *in type* B*.*

Proof. $u^{-1}(u-u^{-1}) = 1-u^{-2} = [K^C(u)]_0^1$ and $u^{-1/2}(u^{1/2}-u^{-1/2}) = 1-u^{-1} = [K^B(u)]_0^1$, which is the first identity; substituting it into (12) gives the second. Both were also checked symbolically (§6). For the loci: $1-u^{-2}$ vanishes at $u=\pm1$ and $1-u^{-1}$ only at $u=+1$; and $q(w_a-w_a^{-1})^2$ vanishes at $w_a=\pm1$ while $q(w_a+w_a^{-1}-2) = q\,w_a^{-1}(w_a-1)^2$ takes the value $-4q \neq 0$ at $w_a=-1$. □

4.1 Why this is a statement and not a resemblance

The identity (13) has to be defended against the objection that a two-variable polynomial of the shape $q(x^{1/2} - x^{-1/2})^2$ can be factored to look like anything. Three coordinates, of which the brief had checked one.

Coordinate 1: the polynomial. Checked by the brief, and it was checked against the wrong paper (§1). With the right entry the match is better, not worse: the discrepancy of a factor u between $u - u^{-1}$ and $1 - u^{-2}$ is the monomial w_a^h of (13), and the brief's reading of $(w_a - w_a^{-1})^2$ as a *square* is correct — the square of one entry at one argument, not a product of two entries at two arguments, and not the kernel-argument doubling of the Pfaffian literature that the brief feared.

Coordinate 2: the slot, the index set, and the count. Motegi's K-matrices act on the $\mathbf{u}$-slot, one per reflected row, with arguments $u_1, \dots, u_n$ (eqs. (5), `eq:reflection` L382, and "$K_a(u)$ denotes the operator acting non-trivially on the a-th component" L388). By (4), $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ *is* the specialisation $\mathbf{u} = \mathbf{w}_{\mathbf{M}}$. Hence:

the K-matrices of $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ are exactly $\{K(w_m) : m \in \mathbf{M}\}$, indexed by the specialisation set $\mathbf{M}$ —

not by the label $\mathbf{I}$, and not by the complement. A long-root collapse deletes $\{a, b\}$ from $\mathbf{M}$ (the hypothesis of Theorem 3 is $\{a, b\} \subseteq \mathbf{M}$), hence deletes exactly two K-matrices, $K(w_a)$ and $K(w_b)$, and ρ^X is by construction the ratio of the two levels. Two rows deleted, two boundary entries, at the two arguments w_a, w_b — and on H these are mutually inverse, $w_b = w_a^{-1}$, which is why one squared entry at one argument suffices.

This is not read off my own algebra. Remark 2 records that Motegi's divisibility theorems make $\prod_{m\in\mathbf{M}}(1 - w_m^{-h}) = \prod_{m\in\mathbf{M}}[K^X(w_m)]_0^1$ an explicit divisor of $\widetilde{WX}_{\mathbf{M},\mathbf{I}}$, with the product running over $\mathbf{M}$ and with the type-correct exponent h in each case. The charge-violating boundary entry, indexed by the K-argument set, is a divisor of the weight function in the source.

Coordinate 3: the disagreement set. Here is the discrimination, and it is why Theorem 3 was worth the retraction. There are two candidate readings:

(U) the *unnormalised* entry $[K(u)]_0^1 = 1 - u^{-2}$ of (1), which is β, γ-free and therefore the same function in both types. Reading (U) predicts a *type-independent* ρ.

(N) the *normalised* entries $[K^X(u)]_0^1$ of (3), which differ between the types by precisely the type B scalar $(1 + u^{-1})^{-1}$. Reading (N) predicts a type-dependent ρ, tracking h.

The two readings *agree* at $w_a = +1$ and *disagree* at $w_a = -1$, which is the only point where the type B scalar $(1 + u^{-1})^{-1}$ is singular. The measurement lands on the disagreement set:

	$[K^C]_0^1$	$\rho^C - 1$	$[K^B]_0^1$	$\rho^B - 1$
$w_a = +1$	0	0	0	0
$w_a = -1$	0	0	2	$-4q \neq 0$

Reading (U) is refuted and reading (N) is confirmed, on a locus where they make opposite predictions. Had [LR]'s type B claim been true, (U) would have been the surviving reading and the identity would have carried no information about the boundary at all — a β, γ-free entry cannot distinguish anything. It is the *error* that supplied the discriminating coordinate.

Remark 7 (the point the brief pre-registered). The brief's Task 2 pre-registered a falsifier at $w_a = w_b = \pm 1$, where $w_a w_b = 1$ and $w_a^h = 1$ both hold, and named the right arm for it: "*the type B WB prefactor* $\prod_k (1 + u_k^{-1})^{-1}$ *has a pole at* $u = -1$ *and is not* $u \mapsto -u$ *invariant, so* $w_a = -1$ *is exactly where B can differ from C — same point, reached from the other arm. Run both signs separately.*" Run both signs separately: that is the table above, and the instruction was exactly right. The brief expected the falsifier to kill the reading; it killed one of the reading's two forms and promoted the other.

Remark 8 (what is still not established). The *occurrence* of $([K^X]_0^1)^2$ in $\rho^X - 1$ is not shown to be *causal*. In the derivation the numerator of ρ^X is $(1-q)^2$, which comes from the bulk factor $(1-q)^{|\mathbf{L}|}$ of the branching formula and contains no K-matrix; the boundary entry appears only after subtracting 1 and restricting to H. So the status of the reading is: polynomial matched, slot and index set and count and type-dependence matched, mechanism unproved. It is mathematics with a named mechanism gap, not a theorem about the lattice model. I record this because the identity is pretty and the temptation to call it an explanation is the exact failure the gate exists to catch.

5 Gap `nonvanH`: a reduction, a stratum, and two failures

The primary target of the session was [LR]'s remaining named hypothesis.

Gap 1 (`nonvanH`)**.** $\mathrm{ord}_{H_{ab}} \widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ *whenever* $\mathbf{M} \geq \mathbf{I}$ *and* $\{a,b\} \cap \mathbf{M} = \emptyset$.

Equivalently, since $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ is a Laurent polynomial (fact (R) of [LR]): $t = w_a w_b - 1$ does not divide $\widetilde{A}_{\mathbf{M},\mathbf{I}}$. It remains open. What follows is a stratum check, a reduction, a closure on part of the domain, and two attempts with their breakdown points.

5.1 Stratum check: nothing is borrowable from the collapse

[LR] Theorem `collapse` has hypothesis $\{a,b\} \subseteq \mathbf{M}$; Gap 1 has hypothesis $\{a,b\} \cap \mathbf{M} = \emptyset$. These strata are *complementary*, so the collapse theorem says nothing whatever on the domain of the gap, and no part of the proof below may borrow from it. The degenerate reading confirms that they must be disjoint: if (9) held with $\{a,b\} \cap \mathbf{M} = \emptyset$ then $\mathbf{M}' = \mathbf{M}$ and it would read $(\rho - 1)\widetilde{A}_{\mathbf{M},\mathbf{I}} \equiv 0 \pmod{t}$, whence $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_H = 0$ since $\rho - 1$ is a unit at H (Proposition 5) — the exact negation of Gap 1. The two statements are about disjoint strata and both can be true; on the overlap they could not be.

For completeness: the third stratum $|\{a,b\} \cap \mathbf{M}| = 1$ is not needed. The hypothesis (**??**) of [LR] Theorem `created` is applied only at $\mathbf{M} = \nu' = \nu \setminus \{a,b\}$, which is disjoint from $\{a,b\}$ by construction; and the stratum $\{a,b\} \subseteq \mathbf{M}$ reduces to the disjoint one by the collapse theorem, since ρ is a unit at H.

5.2 Reduction to $q = 0$

Lemma 9. $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ *is regular at* $q = 0$*, and* $\mathcal{E}^C_{N,n,r}|_{q=0} = 1$*; consequently* $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_{q=0} = F_{N,n,r}(\mathbf{w_M}|\mathbf{w}|\mathbf{I})\big|_{q=0,\ \gamma=0,\beta=1}$.

Proof. Every factor of (6) other than the prefactor has the form $1 - q \cdot (\text{monomial})$, hence equals 1 at $q = 0$; and the prefactor $(1-q)^{-(n-r)}$ equals 1 there too. So $\mathcal{E}^C|_{q=0} = 1$, and in particular $\mathcal{E}^C$ is a unit of the local ring at $q = 0$, so $(\mathcal{E}^C)^{-1}$ is regular there. F is polynomial in q by Motegi's Theorem 3.3 (L491–504; the only denominators are the $u_j - u_k$, free of q). Hence $\widetilde{A} = F \cdot (\mathcal{E}^C)^{-1}$ is regular at $q = 0$ with the stated value. □

Proposition 10 (the reduction)**.** *If* $\widetilde{A}_{\mathbf{M},\mathbf{I}}\big|_{q=0}\big|_H \neq 0$ *then* $\widetilde{A}_{\mathbf{M},\mathbf{I}}\big|_H \neq 0$*. Hence it suffices to prove Gap 1 at* $q = 0$*.*

Proof. By Lemma 9 the two operations $f \mapsto f|_{q=0}$ and $f \mapsto f|_H$ are both defined on $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ and commute (they substitute into disjoint sets of variables, q and w_b). So $\widetilde{A}|_H = 0$ forces $\widetilde{A}|_{q=0}|_H = 0$. □

The reduction is worth having because $q = 0$ collapses the formula almost completely.

Lemma 11 (the $q = 0$ structure)**.** *At* $q = 0$ *the entire dependence of* $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ *on the variables* $\{w_k : k \notin \mathbf{M}\}$ *is through the single product* $\prod_{j=1}^{r}\prod_{k<i_j}(u_j - w_k)$*, with* $\mathbf{u} = \mathbf{w}_{\mathbf{M}}$*. Explicitly,*

$$\widetilde{A}_{\mathbf{M},\mathbf{I}}\big|_{q=0} = \frac{\prod_{k=1}^{n} u_k^{-(N+n)}}{(n-r)!}\sum_{\sigma\in S_n}\sigma\big[\mathcal{B}^0\big], \tag{14}$$

$$\mathcal{B}^0 = \prod_{j=1}^{r}\Big[(u_j^2-1)\,u_j^{N-i_j}\prod_{k<i_j}(u_j-w_k)\Big]\cdot\prod_{j>r}u_j^{N+1}\cdot\prod_{\substack{j<k\\ j\le r}}\frac{-u_k}{u_j-u_k}(u_ju_k-1)\cdot\prod_{r+1\le j<k\le n}u_ju_k. \tag{15}$$

In particular, if $k \notin \mathbf{M}$ *and* $k \geq \max\mathbf{I}$ *then* w_k *does not occur in* $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_{q=0}$*.*

Proof. Lemma 9 reduces the claim to $F|_{q=0}$. In Motegi's Theorem 3.3 the alphabet $\mathbf{w}$ occurs in exactly three places, and two of them die at $q = 0$: $\prod_{k=i_j+1}^{N}(u_j - qw_k) \to u_j^{N-i_j}$ for $j \leq r$, and $\prod_{k=1}^{N}(u_j - qw_k) \to u_j^N$ for $j > r$. The third, $\prod_{k=1}^{i_j-1}(u_j - w_k)$ for $j \leq r$, carries no q and survives unchanged. The remaining factors are $\mathbf{w}$-free: $(1-q)(u_j^2-1) \to (u_j^2-1)$, $\frac{qu_j-u_k}{u_j-u_k} \to \frac{-u_k}{u_j-u_k}$, $(u_ju_k - q) \to u_ju_k$, and $\prod_{j>r}(\beta u_j - \gamma) = \prod_{j>r}u_j$ in type C, which has been absorbed into u_j^{N+1} in (15). This gives (14)–(15). For the last sentence: w_k occurs in $\mathcal{B}^0$ iff $k < i_j$ for some $j \leq r$, i.e. iff $k < \max\mathbf{I}$; and w_k is not among the $\mathbf{u} = \mathbf{w}_{\mathbf{M}}$ because $k \notin \mathbf{M}$. Since the condition is independent of σ, no w_k can appear through the sum either. □

Corollary 12 (the gap closes on a stratum)**.** *Suppose* $\mathbf{M} \geq \mathbf{I}$*,* $\{a, b\} \cap \mathbf{M} = \emptyset$ *and* $\max\mathbf{I} \leq b$*. If* $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_{q=0} \neq 0$*, then* $\operatorname{ord}_{H_{ab}}\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$*.*

Proof. By Lemma 11, w_b does not occur in $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_{q=0}$, so restriction to H — which is the substitution $w_b \mapsto w_a^{-1}$ — acts trivially on it: $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_{q=0}|_H = \widetilde{A}_{\mathbf{M},\mathbf{I}}|_{q=0} \neq 0$. Apply Proposition 10. □

Remark 13 (the residual hypothesis is the wrong way round). Corollary 12 is conditional on $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_{q=0} \neq 0$, which is *stronger* than the converse of triangularity ($\mathbf{M} \geq \mathbf{I} \Rightarrow \widetilde{A}_{\mathbf{M},\mathbf{I}} \neq 0$) that [Q488] already carries as measured, since $\widetilde{A}|_{q=0} \neq 0$ implies $\widetilde{A} \neq 0$ but not conversely. So the corollary trades one unproved hypothesis for a stronger one, and is honest progress only to the extent that the $q = 0$ object is more tractable — which it is, by Lemma 11, and which is also where Motegi identifies $WC|_{q=0}$ with the factorial GQ-functions of Ikeda–Naruse (L555). I have *not* deep-read Ikeda–Naruse and therefore do not cite any nonvanishing statement from it; that is the natural next step and it is a reading task, not a proof task.

How much is the stratum $\max\mathbf{I} \leq b$**?** Counted combinatorially over the domain of Gap 1 (all $a < b$, all $\mathbf{M} \geq \mathbf{I}$ with $\{a, b\} \cap \mathbf{M} = \emptyset$): see §6. As with [LR]'s minimal stratum, the figure is reported at several N and the trend, not the first value, is the claim.

5.3 Two failed attempts, located

Attempt 1: a bidegree corner on $\widetilde{A}$. Refuted by measurement. Write $\widetilde{A}_{\mathbf{M},\mathbf{I}} = \sum_{i,j} c_{ij} w_a^i w_b^j$ with $c_{ij} \in \mathbb{C}(q)[w_k^{\pm 1} : k \neq a, b]$; this is legitimate because $a, b \notin \mathbf{M}$, so H constrains only w_a, w_b. Then $\widetilde{A}|_H = \sum_d \big(\sum_{i-j=d} c_{ij}\big) w_a^d$, and if the maximum of $i - j$ over the support is attained at a *unique* lattice point then the corresponding coefficient survives and $\widetilde{A}|_H \neq 0$. The natural candidate for that point is the corner $(A, B) := (\max_i, \min_j)$.

It is empty. Measured over the whole domain: 0 occupied out of 1 at $N = 2$, 0 out of 12 at $N = 3$, and the same at every case inspected at $N = 4$ (§6). The reason is identifiable. First, I had intended to read the bidegree off Motegi's symmetrisation formula, where each w_k occurs only in linear factors, one per position j, and the positions are not moved by σ — so each σ-term has the *same* bidegree, $\big(n - [a \in \mathbf{I}],\, n - [b \in \mathbf{I}]\big)$. That is true of F and useless for $\widetilde{A}$: the individual σ-terms of $F/\mathcal{E}^C$ are not Laurent polynomials (the denominator $\prod_j (1 - q u_j w_a)$ of $(\mathcal{E}^C)^{-1}$ has w_a-degree n and does not divide any single term), so the bidegree of $\widetilde{A}$ cannot be read term by term. Second, and independently fatal, $\widetilde{A}$ carries the explicit factor $(1 - q w_a w_b)$ — a factor of the $\mathbf{w}$-product in the denominator of (6), present because $a, b \in \mathbf{M}^{\mathrm{c}}$ — whose two monomials 1 and $-q w_a w_b$ have bidegrees $(0, 0)$ and $(1, 1)$ and hence the *same* value of $i - j$. Any such factor ties the maximum, and there is at least one. A corner argument on $\widetilde{A}$ cannot work; it would have to be applied to $\widetilde{A}$ divided by its manifestly H-nonvanishing factors, and I did not find a clean statement of which those are in the unequal-cardinality case.

Attempt 2: the corner at $q = 0$. Reduces to a twisted symmetrisation sum I cannot evaluate. At $q = 0$ the first objection vanishes: by Lemma 9 $\mathcal{E}^C|_{q=0} = 1$, so $\widetilde{A}|_{q=0}$ *is* the symmetrisation sum (14) and the bidegree is σ-uniform. By Lemma 11 the degrees are $A_0 = \#\{i \in \mathbf{I} : i > a\}$ and $B_0 = \#\{i \in \mathbf{I} : i > b\}$, both minimal degree 0. Extracting the corner $(A_0, 0)$ — top in w_a, constant in w_b — the w_a-factors each contribute -1 and the w_b-factors each contribute u_j, giving

$$[w_a^{A_0} w_b^0]\, \widetilde{A}_{\mathbf{M},\mathbf{I}}\big|_{q=0} = \pm \frac{\prod_k u_k^{-(N+n)+2}}{(n-r)!} \sum_{\sigma \in S_n} \sigma\Big[\Big(\prod_{j \le r,\ i_j > a} u_j^{-1}\Big) \mathcal{B}^{0,\flat}\Big], \tag{16}$$

where $\mathcal{B}^{0,\flat}$ is (15) with the columns a, b deleted from every product. When $\mathbf{I} \cap \{a, b\} = \emptyset$, $\mathcal{B}^{0,\flat}$ *is* the bracket (15) for the alphabet $\mathbf{w}^\flat = (w_k)_{k \neq a,b}$ and the relabelled index set $\mathbf{I}^\flat$ (the relabelling is order-preserving, so $\mathbf{M} \geq \mathbf{I}$ gives $\mathbf{M}^\flat \geq \mathbf{I}^\flat$), and the symmetric monomial $\prod_j u_j^2$ pulls out of the sum. **The obstruction is the remaining monomial** $\prod_{j \le r,\, i_j > a} u_j^{-1}$: the set $\{j \le r : i_j > a\}$ is the top A_0 *positions*, not a symmetric function of the u's, so it does not pull out of the σ-sum, and multiplying a position's block by u_j^{-1} does not shift it to another index's block (it would need one fewer $(u - w)$ factor, not one lower power of u). So (16) is a symmetrisation sum twisted by a non-symmetric monomial, and I can neither identify it with a weight function nor show it nonzero. When $\mathbf{I} \cap \{a, b\} \neq \emptyset$ — which does occur in the application, e.g. $N = 3$, $\nu = \{1, 2, 3\}$, $(a, b) = (1, 2)$, $\nu' = \{3\}$, $\lambda = \{1\}$ — there is a second obstruction: the block at the position j with $i_j = a$ becomes a product with *no* hole, which is not an index block of the reduced system at all.

What I would try next. In order: (i) deep-read Ikeda–Naruse on nonvanishing of factorial GQ-functions and see whether it discharges the hypothesis of Corollary 12 (a reading task — and the honest statement is that I currently hold that paper at a level that forbids citing it); (ii) evaluate (16) in the equal-cardinality case $|\mathbf{M}| = |\mathbf{I}|$, where the σ-sum has one term and the whole gap is the type A statement $W_{N,r}(\mathbf{w}_{\mathbf{M}}|\mathbf{w}|\mathbf{I})|_H \neq 0$ ([LR] §`scope` already records this); (iii) look for the

right factorisation $\widetilde{A}$ = (manifestly H-nonvanishing) $\cdot\, Q$ in the unequal-cardinality case, which is what Attempt 1 lacked — Motegi's divisibility theorems (Remark 2) supply such a factorisation at $|\mathbf{M}| = |\mathbf{I}|$, and their $s > r$ analogue is where I would look.

6 Computational verification, with controls and plants

Pure Python + SymPy over $\mathbb{C}(q)(w_1, \ldots, w_N)$ (SageMath is not installed). Scripts and logs: `proofs/code-1017-nonvanH/`. The engine `q484_engine.py` is carried over; `eb_audit.EB` implements (7) and is new.

The normaliser, and that `EB` is right (`eb_audit.py`, `verify_all.py`), with two plants. Lemma 1 verified symbolically for every $\mathbf{M}$ and r at $N = 2, 3$. Then `EB` is checked against Motegi's *independent* theorem `typeBmainproperties`(i) (eq. `normalizationWB` L1386), i.e. against the type B diagonal root-product formula, which is a different theorem from the normalisation factor (7):

	$N = 2$	$N = 3$
`EB` as implemented: match / cases	4/4	8/8
P-A1 inclusive $\mathbf{w}$-diagonal (the type C convention): match	0/4	0/8
P-A2 drop $\prod_j (1 - qu_j)$: match	1/4	—

P-A1 is the sharp plant: it is exactly the error the engine was making, and it fails on every case. P-A2 is *informatively spared* on one case, namely $\mathbf{I} = \emptyset$ with $n = 0$, where its product is empty and the plant is the identity by construction. The two plants sit in different blocks of (7) — the $\mathbf{w}$-diagonal and the linear $\mathbf{u}$-product — which is the point of naming them separately.

The multiplier (`eb_rho.py`, `rhoB_N4.py`), two arms. Over all ($a < b$, $\mathbf{M} \supseteq \{a, b\}$, $\mathbf{I}$), restricted to H, reporting only the *non-degenerate* domain (some side $\neq 0$ on H; a pair with both sides zero cannot distinguish any multiplier):

object tested	domain	ρ^C holds	ρ^B holds
$\widetilde{WC} = WC/\mathcal{E}^C$, $N = 2$	1	1/1	0/1
$\widetilde{WC} = WC/\mathcal{E}^C$, $N = 3$	12	12/12	0/12
$\widetilde{WB} = WB/\mathcal{E}^B$, $N = 2$	1	0/1	1/1
$\widetilde{WB} = WB/\mathcal{E}^B$, $N = 3$	12	**0/12**	**12/12**

Each candidate holds on its own type and fails on the other, and the two candidates disagree on the *whole* domain, so the test is discriminating rather than merely green: it is evaluated on the disagreement set, and the disagreement set is not empty.

Reproducing the retracted number (`rhoB_N4.py`), closing the derivation loop. The proof of Theorem 3 routes through $\Lambda = WB/\mathcal{E}^C$, so it needs [LR]'s measurement as an input rather than as a corroboration. That measurement reproduces: Λ collapses with ρ^C in 1/1 non-degenerate of 4 pairs at $N = 2$ and 12/12 non-degenerate of 48 pairs at $N = 3$ — the same 48 and 12 as [LR] reports, so the retracted number was a correct measurement of the wrong object. Plants in the multiplier block: $\rho \mapsto \rho(1 + q)$ and $\rho \mapsto 1$ both fail 0/12, as required. Hence ρ^B is reached twice independently: by (11) from the reproduced Λ-measurement plus Lemma 1, and by direct

measurement of $\widetilde{WB}$ in the table above. The $N = 4$ arm for $\widetilde{WB}$ had not returned at write-up and is marked *pending*: a claim arm without its plants, or a table without its widest domain, is reported as pending rather than assumed.

The dictionary (symbolic). $\rho^X - 1 - q\, w_a^h([K^X(w_a)]_0^1)^2/\big((1-qw_a^h)(1-qw_a^{-h})\big)$ simplifies to 0 for $X = C$ ($h = 2$) and $X = B$ ($h = 1$); and $w_a^{h/2}[K^X(w_a)]_0^1 - (w_a^{h/2} - w_a^{-h/2}) = 0$ in both. The values in the disagreement table of §4 were computed, not asserted.

Attempt 1, refuted (`corner.py`). Over the whole domain of Gap 1: the bidegree corner $(\max_i, \min_j)$ of $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ in (w_a, w_b) is **occupied in** 0 **cases and empty in** 1 **of** 1 **at** $N = 2$**,** 12 **of** 12 **at** $N = 3$, and likewise at every $N = 4$ case inspected. This is a measurement that refuted a route before any time was spent proving it, and it is recorded as the only outcome of Attempt 1.

Lemma 11 (`q0_struct.py`, `q0b.py`), with a plant, and a pre-registered claim that was wrong. First, $\mathcal{E}^C|_{q=0} = 1$ for every $\mathbf{M}$ and r at $N = 2, 3, 4$. Then the occurrence question. I pre-registered the *iff*: "w_k occurs in $\widetilde{A}_{\mathbf{M},\mathbf{I}}|_{q=0}$ if and only if $k < \max \mathbf{I}$". **The iff is false**: it fails at $N = 3$, $\mathbf{M} = \{2, 3\}$, $\mathbf{I} = \{1, 2\}$, $k = 1$, where w_1 is absent although $1 < \max \mathbf{I} = 2$, and again twice at $N = 4$. The direction Lemma 11 actually asserts and Corollary 12 actually uses — $k \geq \max \mathbf{I} \Rightarrow w_k$ absent — holds in 6/6 at $N = 2$, 27/27 at $N = 3$ and 110/110 at $N = 4$. So the lemma stands as stated and there is additional vanishing beyond it that I did not predict. The **plant** was the same claim with $k \leq \max \mathbf{I}$; it disagrees with the measurement 1 time at $N = 2$ and 7 times at $N = 3$, which is what shows the two thresholds are distinguishable by this scan at all (8 disagreements at $N = 4$). Recording this separately matters: a pre-registered *iff* that fails in its harmless direction is a bookkeeping error in the prediction, not in the lemma, and collapsing the two directions would have reported either a false lemma or a false green.

The stratum of Corollary 12 (`q0b.py`). Counted combinatorially over all ($a < b$, $\mathbf{M} \geq \mathbf{I}$, $\{a, b\} \cap \mathbf{M} = \emptyset$):

N	2	3	4	5	6	7	8
domain	1	12	94	610	3558	19376	100588
$\max \mathbf{I} \leq b$	1	11	77	444	2306	11241	52573
%	100.0	91.7	81.9	72.8	64.8	58.0	52.3

Decreasing, and reported over seven consecutive N rather than from its head: [LR]'s minimal stratum went 100% → 92% → 78% → 61% → 44% and its first two values read as a closure. Corollary 12 is a partial closure on a stratum that is plausibly asymptotically small — the figures are consistent with a limit at or below $\frac{1}{2}$ and I make no claim about the limit — so it does not reduce Gap 1 to a boundary case.

7 Scope

Retracted. [LR] Theorem `collapse`, final sentence ("the same ρ holds in type B") and its 48/48; [Q484] Theorem A, type B arm (48/48). Both measured $WB/\mathcal{E}^C$. The type B *statement* of Theorem 3 replaces the former; the latter awaits re-measurement.

Unaffected. Everything in [LR] and [Q484] stated for type C. $\mathcal{E}^C$ is the correct normaliser for type C and the engine's type C arm is checked against Motegi's independent diagonal root-product formula. In particular [LR] Theorems `easy`, `simple`, `recursion`, `created` and Proposition `rho` stand as type C statements — but [LR]'s remark that they "hold in type B to whatever extent those facts do" must now be read with Theorem 3: the facts (T), (R), (D), (S), (A) have type B analogues with the short-root substitution, and the type B versions of those theorems carry ρ^B, not ρ^C. I have not worked them out.

Open. Gap 1 (`nonvanH`), now with a reduction to $q = 0$ (Proposition 10), an explicit $q = 0$ structure (Lemma 11), a conditional closure on $\max \mathbf{I} \leq b$ (Corollary 12), and two routes closed off with reasons (§5.3). [LR] Gap `prop` (`propagation`), untouched, and still the bulk of the problem. Coordinate 3 of the boundary dictionary (Remark 8): the mechanism.

Q500. The brief gated the boundary reading on a slot check with three pre-registered outcomes. The verdict is **0A, same slot** — the K-matrices of $\widetilde{A}_{\mathbf{M},\mathbf{I}}$ are indexed by the specialisation set $\mathbf{M}$, a collapse deletes exactly two of them, and the charge-violating entry is a divisor of the weight function in Motegi's own divisibility theorems — reached by refuting the brief's own version of the claim (wrong paper, wrong entry, and a ρ that turned out to be type-dependent). It is promoted from shape to mathematics with a named mechanism gap, not to a theorem.

Source hygiene

Every fact attributed to Motegi is from arXiv:2610.15344, read first-hand in the e-print source `typebcweightfunctions_arxivversionfinal-3.tex`, with line numbers given at the point of use: (1) L332–345; (3) and (2) L352–375; (5) L489–505; (6) L1051–1062; (7) L1361–1367; the two divisibility theorems L1086–1089 and L1394–1402; the type B diagonal formula L1386; the symmetrisation formula (Motegi Thm 3.3) L491–504; the $q = 0$ identification with factorial GQ/GB-functions L555. The K-matrix of arXiv:1703.07924 is quoted from `arxiv-1703.07924-source.TEX` L300–321, and is quoted only in order to record that it is *not* the K-matrix of the paper under discussion. Ikeda–Naruse is named in Remark 13 as a reading task and no claim rests on it. No claim above rests on a document whose author is also its subject.